

Unsupervised machine learning for anomaly detection in private 5G core networks to secure industrial IoT operations.

Manufacturing × Threat detection and response × Machine learning · OS001 v3 · refreshed 2026-08-20

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
57 is the world moving	74 can we play, can we win	MEDIUM 0 named in the evidence	€447m observed evidence · per year	LATER research and standards activi...	L1 28 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

This brief was written against an earlier version of the topic. The evidence has moved since; regenerate before using it with a customer.

The opportunity

This opportunity is about deploying privacy-preserving federated learning for intrusion detection in smart factory IIoT networks, enabling real-time anomaly detection without exposing sensitive production data. It targets manufacturing customers with distributed IIoT environments who need to detect cyber threats while complying with data privacy and sovereignty requirements. The timing is driven by the convergence of mature federated learning research and the urgent need for adaptive, privacy-preserving security in industrial settings.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€623m €64.5m – €4.9bn	€447m €46.2m – €3.5bn	€21.5m €2.2m – €167m	observed
Observed: tendered contract value, annualised	€50.5m €50.5m – €184m	€50.5m €50.5m – €184m	€2.4m €2.4m – €8.8m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Machine learning	3.3 % of enterprises (10+ employees)	observed	Eurostat · isoc_eb_ain2 · E_AI_TML	2025
Annual value of one engagement	€488k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	4.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 1 of 1 declared geographies are outside the European reference data (SA); the estimate covers the European footprint only.

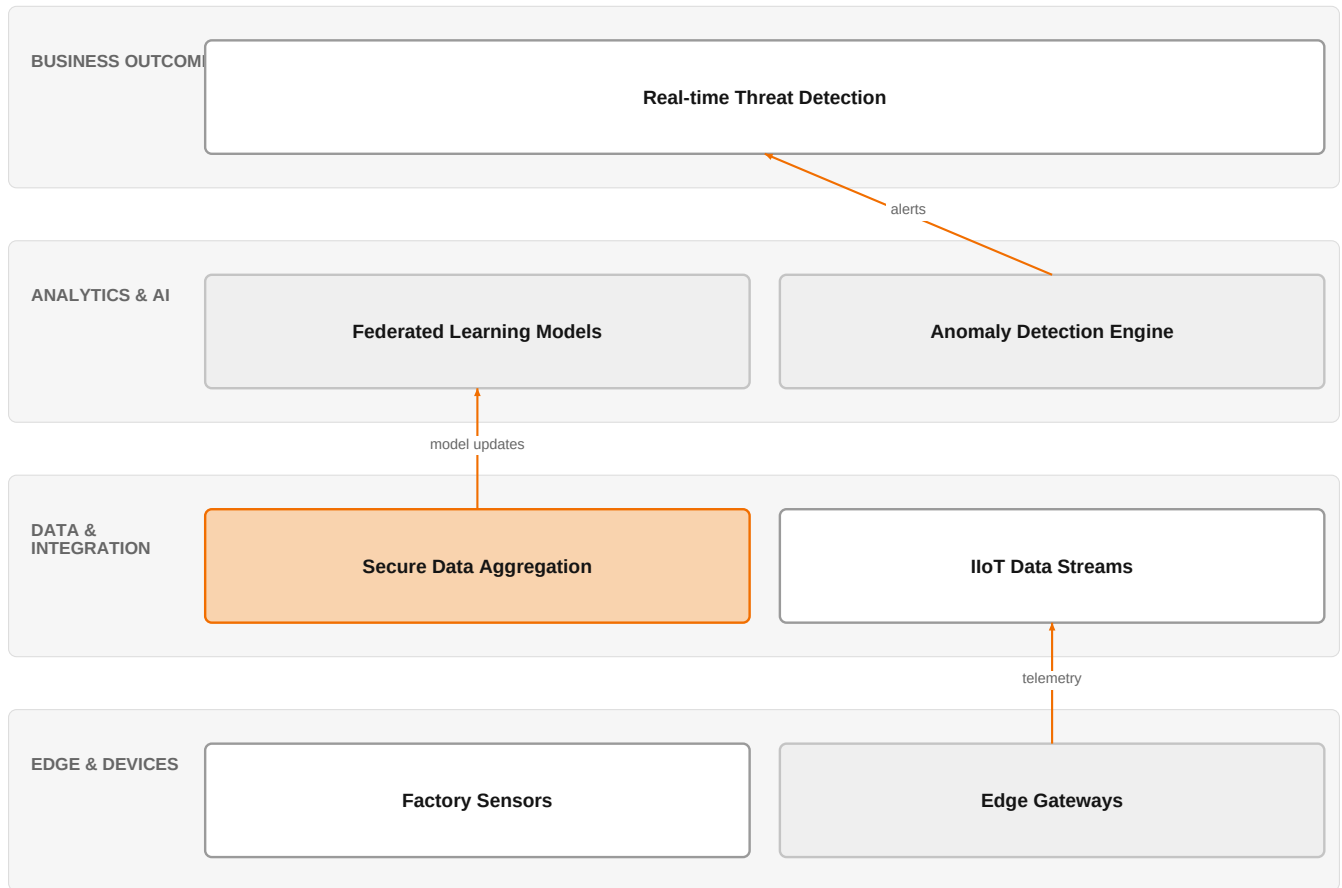
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€50.5m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	4.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 16 matching notices, matched on use case via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 1 of 1 declared geographies are outside the European reference data (SA); the estimate covers the European footprint only.

The solution, in one picture

Federated IIoT Anomaly Detection



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how federated learning enables privacy-preserving anomaly detection across distributed factory sensors, with Orange providing the AI/security expertise and secure aggregation via partners.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The market is shifting as federated learning frameworks for industrial anomaly detection have matured, with systematic reviews and multiple frameworks (e.g., SecuFL-IoT, ROCHE) demonstrating adaptive, privacy-preserving capabilities. Traditional centralized intrusion detection systems are failing against zero-day attacks and lack scalability for high-volume IIoT traffic, while privacy concerns prevent sharing sensitive production data. Research shows a clear trend toward decentralized, collaborative models that preserve data privacy while enabling real-time threat detection.

Sources: SIG-C6ED90595BBC [openalex.org 2026-04-18](https://openalex.org/2026-04-18); SIG-F633044AF87A [openalex.org 2026-01-29](https://openalex.org/2026-01-29); SIG-CF12E7BF5186 [openalex.org 2025-09-22](https://openalex.org/2025-09-22); SIG-8EC0D797B8D2 [openalex.org 2025-10-27](https://openalex.org/2025-10-27); SIG-CEE282CB905A [openalex.org 2025-09-08](https://openalex.org/2025-09-08)

Who buys, and why now

The buyer is typically the Chief Information Security Officer (CISO) or Head of Industrial Cybersecurity at a manufacturing firm, while the operational pain is felt by plant managers and OT security teams who face increasing cyber threats and the challenge of detecting anomalies without disrupting production or exposing proprietary process data. The trigger is often a security incident, an audit finding, or a regulatory pressure to improve threat detection while maintaining data sovereignty.

Why it is hot now — every claim bound to a dated source

- Federated learning frameworks enable adaptive and privacy-preserving anomaly detection in smart industrial networks. [SIG-F633044AF87A]
- Federated learning is used for intrusion detection in IIoT networks, preserving privacy by minimizing raw data transmission. [SIG-3AC93BC5D991]
- Federated learning is explored for intrusion detection in IIoT networks, enabling privacy-preserving security analytics. [SIG-3AC93BC5D991]
- Industrial control systems face dual risks in cyber and physical domains, requiring cross-domain risk assessment. [SIG-005CE36DA1FC]
- Wide-bandgap technologies introduce unique sensor corruption and cybersecurity risks in ICS, addressed by ML-assisted anomaly detection. [SIG-8045366576FF]
- Unsupervised CoAP-based anomaly detection in private 5G core networks can identify security threats in industrial IoT without labeled data. [SIG-D2558A0D8EDD]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution leveraging its Cybersecurity experts and AI/Data/Cloud experts to design and deploy a federated learning-based intrusion detection system. The engagement would start with an assessment of the customer's IIoT architecture and data flows, then build a proof-of-concept using Orange's managed services and possibly partner cloud infrastructure (e.g., AWS or Google Cloud) for secure aggregation. The solution would be certified against ISO 27001 and TISAX where relevant, and could integrate with Orange Cyberdefense managed services for ongoing monitoring and response.

Why Orange can win

Orange can win by combining its deep cybersecurity expertise with its ability to deliver sovereign, certified solutions (ISO 27001, TISAX) that address data privacy concerns. The presence of Orange Group researchers and defense/security experts adds credibility in advanced AI and security. However, the assets are thin in terms of a specific federated learning product, so Orange would need to partner with cloud providers or leverage its managed services to assemble a credible offering.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Orange Drone Guardian	offer	L1	offer provides the technology in a matching domain	unconfirmed
Orange Cyberdefense managed services	offer	L1	offer addresses the use case but not this technology	unconfirmed
AWS	partner	L2	partner provides the required technology	unconfirmed
Google Cloud	partner	L2	partner provides the required technology	unconfirmed
NiCE	partner	L2	partner provides the required technology	unconfirmed
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Cybersecurity experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed

Saint-Gobain Glass	reference	SUP	published reference in this vertical, different use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

MEDIUM competitive intensity (score 7.1; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include hyperscalers AWS and Google Cloud, which offer machine learning platforms but may lack deep industrial OT security expertise. Atos/Eviden competes in both machine learning and cybersecurity, potentially offering integrated solutions. NiCE, a customer-experience platform, is also an Orange partner and sells machine learning. Data and AI platforms like Databricks, Dataiku, and Palantir offer advanced analytics but may not have industrial-specific security focus. Accenture is active in manufacturing and could bring systems integration capabilities. The customer will compare Orange's ability to combine security, sovereignty, and industrial domain knowledge against these alternatives.

Sources: SIG-F633044AF87A openalex.org 2026-01-29; SIG-CF12E7BF5186 openalex.org 2025-09-22

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells Machine learning — also an Orange partner	structural	—
Google Cloud	Hyperscaler	sells Machine learning — also an Orange partner	structural	—
Atos / Eviden	Global systems integrator	sells Machine learning; competes in Cybersecurity	structural	—
NiCE	Customer-experience platform	sells Machine learning — also an Orange partner	structural	—
Databricks	Data and AI platform	sells Machine learning	structural	—
Dataiku	Data and AI platform	sells Machine learning	structural	—
Palantir	Data and AI platform	sells Machine learning	structural	—
Accenture	Global systems integrator	active in Manufacturing	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Competitor analysis

7 of 8 competitors on this space have been read from their own published pages; 1 could not be read and are marked below.

The shape of the field

The field for anomaly detection in private 5G core networks for industrial IoT is populated by hyperscalers (AWS, Google Cloud) offering general-purpose cloud and AI services, global systems integrators (Atos/Eviden, Accenture) providing consulting and integration, and data/AI platforms (Databricks, Dataiku) focusing on analytics and machine learning. NiCE is a customer-experience platform with no direct relevance, and Palantir's position is unreadable. The gap lies in a vertically integrated, sovereign solution that combines network infrastructure, security, and machine learning, which Orange is uniquely positioned to fill with its owned network and cybersecurity expertise.

AWS Hyperscaler · also an Orange partner · structural presence

AWS presents itself as a comprehensive cloud and AI provider with a broad portfolio of services, solutions, and guidance for various industries and technical use cases. It targets organizations from startups to enterprises, offering pre-vetted architectures and partner solutions to accelerate cloud adoption and innovation.

Doing here: AWS provides a broad portfolio of cloud and AI services, including security, identity, and compliance solutions to automate threat detection and response. It offers IoT solutions to connect, ingest, process, and analyze device data with multi-layered security. AWS also provides pre-vetted Solutions packages that combine cloud services and configurations for specific industry needs, vetted by AWS architects for reliability, security, and cost-efficiency.

How Orange differentiates: Orange differentiates by offering a sovereign, EU-resident private 5G core network with integrated anomaly detection, leveraging Orange Cyberdefense managed services and ISO 27001 certification. Unlike AWS's generic cloud services, Orange provides a fully managed, end-to-end solution that includes the network infrastructure, ensuring data residency and compliance with European regulations. Orange's deep expertise in telecommunications and cybersecurity, backed by Orange Group researchers, enables a more tailored and secure solution for industrial IoT operations. While AWS offers scalable cloud infrastructure, Orange's strength lies in its ability to deliver a vertically integrated, sovereign solution that meets the stringent security and regulatory requirements of European manufacturers.

Anchored on: Orange Cyberdefense managed services · ISO 27001 · Orange Group researchers

What they do better: AWS offers a vast ecosystem of cloud services and pre-vetted solutions that provide scalability and flexibility, which Orange may not match in breadth.

Google Cloud Hyperscaler · also an Orange partner · structural presence

Google Cloud presents itself as a comprehensive cloud platform for building, deploying, and scaling applications, with a strong emphasis on AI, data analytics, and security. It offers a wide range of services including infrastructure, data cloud, and AI solutions, targeting enterprises and developers.

Doing here: Google Cloud provides a comprehensive cloud platform for building, deploying, and scaling applications, with a strong emphasis on AI and data analytics. It introduces Google AI Threat Defense to help customers outpace adversaries and offers Gemini Enterprise for agent development and orchestration. Google Cloud also provides an agentic Data Cloud that powers the system of action.

How Orange differentiates: Orange differentiates by offering a private 5G core network with built-in anomaly detection, ensuring data sovereignty and compliance with EU regulations, which is critical for industrial IoT. Unlike Google Cloud's general-purpose AI and threat defense, Orange provides a specialized solution that integrates network security with machine learning, backed by Orange Cyberdefense managed services and ISO 27001 certification. Orange's ownership of the network infrastructure allows for deeper integration and real-time threat response, which is not possible with a hyperscaler's cloud-only approach. While Google Cloud offers advanced AI capabilities, Orange's strength lies in its ability to deliver a sovereign, end-to-end solution tailored to the specific needs of European manufacturers.

Anchored on: Orange Cyberdefense managed services · ISO 27001

What they do better: Google Cloud has advanced AI and data analytics capabilities that are broader and more mature than Orange's, particularly in generative AI and large-scale data processing.

Atos / Eviden Global systems integrator · structural presence

Eviden, the product division of Atos Group, presents itself as a provider of mission-critical systems, sovereign cybersecurity products, and AI-powered video analytics for public and private organizations. It emphasizes securing people, data, and critical infrastructure across digital and physical domains, with a focus on digital sovereignty and resilience.

Doing here: Eviden provides highly certified software and hardware cybersecurity products for data protection, access management, and identity security. It offers a Digital Sovereignty offering to help organizations retain control over their digital assets. Eviden delivers mission-critical communication solutions including private 4G LTE/5G networks for

governments, defense, and offshore wind. It also provides network encryption solutions certified Common Criteria EAL4+ and ANSSI QS.

How Orange differentiates: Orange differentiates by offering a comprehensive managed service that combines private 5G core networks with unsupervised machine learning for anomaly detection, backed by Orange Cyberdefense managed services and ISO 27001 certification. Unlike Eviden, which provides cybersecurity products and private network solutions separately, Orange integrates these into a single, end-to-end offering, ensuring seamless security and operational efficiency. Orange's expertise in both telecommunications and cybersecurity, along with its research capabilities, enables a more holistic approach to securing industrial IoT. While Eviden has strong sovereign cybersecurity products, Orange's advantage lies in its ability to provide a fully managed, integrated solution that reduces complexity for manufacturers.

Anchored on: Orange Cyberdefense managed services · ISO 27001 · Orange Group researchers

What they do better: Eviden has a strong portfolio of certified cybersecurity products and private network solutions, with deep expertise in defense and government sectors, which may be more specialized than Orange's offerings.

NiCE Customer-experience platform · also an Orange partner · structural presence

NiCE presents itself as a provider of AI-powered customer experience (CX) platforms, primarily for contact centers, offering a suite of solutions for self-service, agent assist, workforce management, and analytics. It targets enterprises and SMBs across various industries, emphasizing scalability, cloud deployment, and integration.

Doing here: NiCE provides AI-powered customer experience platforms, primarily for contact centers, with solutions for self-service, agent assist, workforce management, and analytics. It offers CXone, a cloud-based call center software with advanced routing and quality assurance. NiCE also provides real-time agent assist solutions with live AI-powered guidance, including transcription and intent detection.

How Orange differentiates: Orange differentiates by focusing on industrial IoT security with private 5G core networks and anomaly detection, a domain where NiCE has no presence. NiCE's expertise is in customer experience and contact center solutions, which are unrelated to the security of industrial networks. Orange leverages its telecommunications infrastructure and cybersecurity expertise, including Orange Cyberdefense managed services and ISO 27001 certification, to deliver a specialized solution for manufacturers. While NiCE offers AI-driven customer engagement, Orange's value proposition is in securing critical industrial operations, a fundamentally different and more specialized market.

Anchored on: Orange Cyberdefense managed services · ISO 27001

What they do better: NiCE has a strong platform for AI-driven customer experience, with advanced capabilities in contact center automation and analytics, which Orange does not offer.

Databricks Data and AI platform · structural presence

Databricks presents itself as a unified data, analytics, and AI platform that enables enterprises to build and deploy AI agents, applications, and analytics on a single governed foundation. It targets a broad range of industries, including manufacturing, financial services, public sector, energy, healthcare, and more, with a focus on eliminating data silos and enabling production-grade AI.

Doing here: Databricks provides a unified platform for data, analytics, and AI, enabling users to build agents, apps, and natural language insights. It offers Lakebase, a serverless Postgres database, and Agent Bricks for building production-ready AI agents. Databricks also provides Unity Catalog for unified governance of data, models, dashboards, and agents, and Lakeflow for building reliable ETL pipelines.

How Orange differentiates: Orange differentiates by offering a complete, managed solution that includes the private 5G core network itself, with integrated anomaly detection using unsupervised machine learning. Unlike Databricks, which provides a data and AI platform, Orange owns the network infrastructure, enabling real-time threat detection and response at the network level. Orange's solution is backed by Orange Cyberdefense managed services and ISO 27001 certification, ensuring a secure and compliant environment for industrial IoT. While Databricks offers powerful data analytics and AI capabilities, Orange's strength lies in its ability to provide a vertically integrated, end-to-end security solution that is tailored to the specific needs of manufacturers.

Anchored on: Orange Cyberdefense managed services · ISO 27001

What they do better: Databricks has a robust data and AI platform with advanced analytics and governance capabilities, which may be more flexible for custom machine learning models than Orange's integrated approach.

Dataiku Data and AI platform · structural presence

Dataiku presents itself as an enterprise AI platform that unifies data preparation, machine learning, generative AI, and AI agents under a single governance framework. It targets large organizations across industries, emphasizing collaboration between technical and non-technical users, orchestration across any infrastructure, and governance to ensure compliance and risk management.

Doing here: Dataiku provides a universal AI platform that combines data preparation, machine learning, generative AI, and AI agents in one governed system. It offers pre-built solutions for specific industries, including manufacturing, and provides Dataiku Govern for centralized AI governance. Dataiku also offers the LLM Mesh as a secure gateway to connect to thousands of LLMs and vector stores.

How Orange differentiates: Orange differentiates by offering a specialized solution for anomaly detection in private 5G core networks, leveraging its telecommunications infrastructure and cybersecurity expertise. Unlike Dataiku, which provides a general-purpose AI platform, Orange's solution is deeply integrated with the network, enabling real-time threat detection and response. Orange's offering is backed by Orange Cyberdefense managed services and ISO 27001 certification, ensuring a secure and compliant environment for industrial IoT. While Dataiku offers a flexible AI platform with strong governance, Orange's advantage lies in its ability to provide a turnkey, network-centric security solution that is specifically designed for the unique challenges of industrial IoT.

Anchored on: Orange Cyberdefense managed services · ISO 27001

What they do better: Dataiku has a comprehensive AI platform with strong governance and collaboration features, which may be more adaptable for various AI use cases beyond anomaly detection.

Palantir Data and AI platform · structural presence

Their published position is unread — 40 page(s) fetched, none carried readable text — the site renders its content client-side. Nothing below is inferred on their behalf.

Doing here: Palantir's published position is unreadable because their site renders content client-side, and no readable text was extracted from the fetched pages. Therefore, their activity in this space cannot be assessed from the provided material.

How Orange differentiates: Given that Palantir's published position is unreadable, Orange differentiates by offering a transparent, documented solution for anomaly detection in private 5G core networks, backed by Orange Cyberdefense managed services and ISO 27001 certification. Orange's solution is built on its owned network infrastructure, providing a level of integration and control that is difficult to match. While Palantir is known for its data integration and analysis platforms, Orange's strength lies in its ability to deliver a sovereign, end-to-end security solution for industrial IoT, with clear documentation and support.

Anchored on: Orange Cyberdefense managed services · ISO 27001

What they do better: Palantir has a strong reputation for data integration and analysis in complex environments, which may be valuable for certain use cases, but its lack of transparent documentation makes it difficult to compare directly.

Accenture Global systems integrator · structural presence

Accenture presents itself as a global professional services company that helps organizations across industries reinvent their operations through strategy, consulting, technology, and platform solutions. It emphasizes digital transformation, AI, and cloud to drive growth and resilience for clients in both private and public sectors.

Doing here: Accenture offers connected supply chain solutions that provide real-time visibility, control, and cost savings through integrated platforms. It partners with Blue Yonder, Coupa, Kinaxis, and O9 for supply chain solutions. Accenture also provides consulting services to public sector organizations to help them build resilient operations and improve citizen experiences.

How Orange differentiates: Orange differentiates by offering a specialized, technology-centric solution for anomaly detection in private 5G core networks, leveraging its owned network infrastructure and cybersecurity expertise. Unlike Accenture, which provides consulting and integration services, Orange delivers a managed service that includes the network, the anomaly detection, and the response, backed by Orange Cyberdefense managed services and ISO 27001 certification. Orange's solution is more directly aligned with the security needs of industrial IoT, providing a turnkey approach that reduces the need for extensive integration projects. While Accenture offers broad consulting and supply chain expertise, Orange's advantage lies in its ability to provide a deep, network-level security solution that is essential for protecting critical industrial operations.

Anchored on: Orange Cyberdefense managed services · ISO 27001

What they do better: Accenture has extensive consulting and integration capabilities across industries, which may be valuable for large-scale transformation projects, but it lacks the network infrastructure and specialized security focus that Orange brings.

Qualifying questions for the first meeting

- 1 Do you currently have any intrusion detection systems in place for your IIoT networks, and if so, how do they handle data privacy?

- 2 Have you experienced any security incidents or near-misses that were difficult to detect due to data silos or privacy constraints?
- 3 What are your main concerns about sharing production data with a central security analytics platform?
- 4 Are you exploring federated learning or other privacy-preserving AI techniques for your industrial security?
- 5 Do you have a distributed IIoT environment across multiple sites that would benefit from collaborative threat detection?

Objections you will meet

They will say	You can answer
Federated learning is too immature for production use in our factory.	You're right that it's still emerging, but the research is converging on practical frameworks. We can start with a proof-of-concept on a limited set of sensors to validate performance and then scale gradually.
We already have a security solution from a hyperscaler.	Hyperscalers offer strong ML platforms, but they may not have deep OT security expertise or the sovereign, certified infrastructure that Orange can provide. We can complement your existing tools with a federated layer that respects data privacy.
We're concerned about the cost and complexity of implementing this.	That's a valid concern. We can design a phased approach, starting with a small pilot to demonstrate value, and we can leverage our managed services to reduce operational burden.

Risks and unknowns

The main risk is that federated learning for IIoT intrusion detection is still largely in research phase, with limited production deployments. There are unknowns about the practical performance in real factory environments, the overhead of secure aggregation, and the ability to integrate with existing OT systems. Additionally, the competitive landscape is crowded, and Orange's lack of a proprietary federated learning platform could be a disadvantage unless partnerships are leveraged effectively.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a manufacturing prospect, highlighting Orange's expertise in AI/Data/Cloud and Cybersecurity, and referencing our work with Colgate-Palmolive, Mondelez International, and Saint-Gobain Glass to illustrate our understanding of industrial environments, while proposing a federated learning-based intrusion detection pilot that respects data privacy.
Sales	In a conversation with a manufacturing client's IT or OT security leader, mention that Orange Cyberdefense managed services, backed by ISO 27001 and TISAX certifications, can help them explore privacy-preserving threat detection using federated learning, without exposing their raw operational data.
Strategist / Innovator	Commission a deep-dive brief on federated learning for IIoT intrusion detection, focusing on how Orange's AI/Data/Cloud and Cybersecurity experts can leverage partnerships with AWS, Google Cloud, and NVIDIA to build a privacy-preserving anomaly detection capability for manufacturing clients.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-D2558A0D8EDD	2026-08-01	Computer Communications	1	Unsupervised CoAP-based anomaly detection in private 5G core network for Industrial-IoT
SIG-C6ED90595BBC	2026-04-18	openalex.org	1	Towards collaborative and privacy-preserving industrial time-series anomaly detection with fede...
SIG-6B094A3F3A46	2026-03-26	openalex.org	1	Sybil attack defense in blockchain-based industrial IoT systems using decentralized federated l...
SIG-36DA075F2A6A	2026-03-02	openalex.org	1	IoT-based Industrial Safety and Monitoring System with Real-Time Alert Mechanism
SIG-F633044AF87A	2026-01-29	openalex.org	1	SecuFL-IoT: an adaptive privacy-preserving federated learning framework for anomaly detection i...
SIG-DA32C362F5E9	2026-01-12	openalex.org	1	FedQmc: A Federated Learning Backdoor Defense Method Based on Quantum State Encoding and Multis...
SIG-C201BF32AE96	2026-01-07	openalex.org	1	Federated Learning for Secure Industrial Automation and Grid Optimization
SIG-005CE36DA1FC	2026-01-01	Elsevier BV	1	A quantitative risk assessment method for industrial control system safety and security
SIG-90064685656B	2025-12-23	openalex.org	1	HierTFL: Hierarchical Scheduling for Industrial TSN-Enhanced Federated Learning System
SIG-145E4A264E5D	2025-12-22	openalex.org	1	Task-Oriented Network Reliability for Federated Learning-Enabled Industrial Internet of Things
SIG-3AC93BC5D991	2025-12-19	openalex.org	1	Federated Learning-Based Intrusion Detection in Industrial IoT Networks
SIG-AECDD651DA98	2025-12-17	openalex.org	1	Edge-AI integrated secure wireless IoT architecture for real time healthcare monitoring and fed...
SIG-1C479A2B44CD	2025-11-01	openalex.org	1	CoperFed: A covert personalized federated learning framework for Industrial Control Systems int...
SIG-904D344AF0C0	2025-10-28	openalex.org	1	Federated learning based on two-stage knowledge distillation for intrusion detection in industr...
SIG-C8282154F431	2025-10-28	openalex.org	1	Lightweight Signal Processing and Edge AI for Real-Time Anomaly Detection in IoT Sensor Networks
SIG-8EC0D797B8D2	2025-10-27	openalex.org	1	AI-Powered Security for IoT Ecosystems: A Hybrid Deep Learning Approach to Anomaly Detection
SIG-98BCC6F19BE4	2025-10-27	openalex.org	1	QuantumTrust-FedChain: A Blockchain-Aware Quantum-Tuned Federated Learning System for Cyber-Res...
SIG-F48074B55B20	2025-10-16	openalex.org	1	Implementation of edge AI for early fault detection in IoT networks: evaluation of performance ...
SIG-A49A291BF70B	2025-10-01	openalex.org	1	PureChain-enhanced federated learning for dynamic fault tolerance and attack detection in distr...
SIG-CF12E7BF5186	2025-09-22	openalex.org	1	ROCHE: A Robust and End-to-End Privacy-Preserving Federated Learning Framework for Intrusion De...
SIG-B75BEF75986F	2025-09-08	openalex.org	1	Trustworthy Adaptive AI for Real-Time Intrusion Detection in Industrial IoT Security
SIG-CEE282CB905A	2025-09-08	openalex.org	1	Survey of Federated Learning for Cyber Threat Intelligence in Industrial IoT: Techniques, Appli...
SIG-5BF0E6E15A2F	2025-08-29	openalex.org	1	Decentralized AI on The Edge: Implementing Federated Learning for Predictive Maintenance in Ind...
SIG-8FD8EE65E437	2025-08-19	openalex.org	1	Differential privacy-enabled federated learning for secure neural synchronization in protecting...
SIG-565963A66A79	2026-06-01	arxiv.org	3	IstGPT: LLM-based Anomaly Detection for Spatial-Temporal Graph in Industrial Systems
SIG-2424954F4021	2026-05-24	arxiv.org	3	ASTRO: Adaptive Spatio-Temporal Reinforcement Optimization for GNN Powered Anomaly Detection in ...

SIG-A06EDB8B4416	2026-04-22	arxiv.org	3	IoT-Enhanced CNN-Based Labelled Crack Detection for Additive Manufacturing Image Annotation in ...
SIG-8045366576FF	2025-10-25	arxiv.org	3	Adapting Noise-Driven PUF and AI for Secure WBG ICS: A Proof-of-Concept Study

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-20T10:10:31+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:40:27+00:00
Pipeline	0.1.0	2026-08-20

Generated 2026-08-20 13:57 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.